

Multipartite Entanglement via Quantum Fisher Information (QFI) in 1D Heisenberg Model

What QFI captures, where it helps, and where it fails

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Problem Statement and Motivation

- Study of quantum systems: **entanglement**
- Low dim. quantum systems - **strongly correlated**, and **highly entangled**
- Spin models strongly coupled by exchange interactions - a suitable platform
- **Aim:** Study multipartite entanglement in 1D Heisenberg Model



- Background informs approach: the usual suspects, the need for more.
- Quantum Fisher Information (QFI) as a candidate: a brief overview.
- Operational setup using QFI, results and implications for entanglement in the target model.
- Conclusions and limitations.
- Open directions and future outlook.



The need for more...

- Entanglement quantification is generally difficult, and comes with a lot of caveats.
- ... Here: From "our knowledge of entanglement theory", we know what works, what does not and why.
- In the context of 1D Heisenberg type systems, what are our options?
 - Concurrence, Tangles and *QFI*.



Why/Why not?

Witness	Remarks
Concurrence $C(\rho) = \max(0, 2\lambda_{\max} - \text{tr}(R))$ $R = \sqrt{\sqrt{\rho}\tilde{\rho}\sqrt{\rho}}$	No widely accepted definition of concurrence for systems (larger quantum objects) other than a pair of qubits.
Tangles $\tau_1 = 1 - 4 \sum_{\alpha} \langle S^{\alpha} \rangle^2$ $\tau_2 = 2 \sum_{r \neq 0} C_r^2$	Size-based limits on information prevent access to genuine multipartite entanglement.
QFI	Operational, scalable as witness.

This motivated the focus on QFI as the main diagnostic.



QFI as a witness

- We can define quantum Fisher information $F(\rho, A)$ as [2]

$$\mathbf{F}(\rho, \mathbf{A}) = \frac{1}{4} \text{tr} \rho \mathbf{L}^2 \quad ; \quad \mathbf{Var}(\mathbf{A})_\rho \geq \frac{1}{\mathbf{F}(\rho, \mathbf{A})} \quad (1)$$

- For a thermal ensemble ρ , QFI can be formulated as [11],

$$\mathbf{F}_Q(\mathbf{T}) = \frac{4}{\pi} \int_0^\infty \tanh\left(\frac{\omega}{2\mathbf{T}}\right) \chi''(\omega, \mathbf{T}) d\omega \quad (2)$$

- If ρ achieves a sufficiently large QFI to break classical bounds,

$$f_Q > m(h_{\max} - h_{\min})^2 \quad ; \quad m \in \mathbb{Z}_+ \quad \text{s.t.} \quad m \mid N \quad (3)$$

with f_Q being the QFI density, then ρ **must** be **$m+1$ -partite entangled** [12].

For $h_{\max} = 1/2$ and $h_{\min} = -1/2$ we need,

$$f_Q > m \quad ; \quad m \in \mathbb{Z}_+ \quad \text{s.t.} \quad m \mid N$$



Model, observable, and workflow

- We have,

$$\hat{H} = -J \sum_{\langle i,j \rangle}^N [S_i^z S_j^z + \frac{1}{2}(S_i^+ S_j^- + S_i^- S_j^+)] + h_z \sum_i S_i^z \quad (5)$$

- For QFI we **choose the local observable**: $\hat{\mathcal{O}}_i = \mathbf{S}_i^z$
- Corresponding susceptibility:

$$\chi''(q, \omega) = \sum_{i,j} e^{iq \cdot (i-j)} \sum_{\lambda, \lambda'} (p_\lambda - p_{\lambda'}) \langle \lambda | \hat{S}_i^z | \lambda' \rangle \langle \lambda' | \hat{S}_j^z | \lambda \rangle \\ \times \pi \delta(\omega - E_{\lambda'} + E_\lambda)$$

where, $p_\lambda = e^{-\beta E_\lambda} / Z$; $Z = \sum_\lambda e^{-\beta E_\lambda}$.



Analytical benchmark for $N = 2, B = 0$

$$f_Q(T) = 2 \frac{(e^{3\beta J/4} - e^{-\beta J/4})}{(e^{3\beta J/4} + 3e^{-\beta J/4})} \tanh \frac{\beta J}{2} \quad (6)$$

PPT Criterion

-ve e.values
for $\rho_{AB}^{T_B}$
 $\Downarrow$
Entangled!

Benchmark confirms
entangled regime; sanity
check.

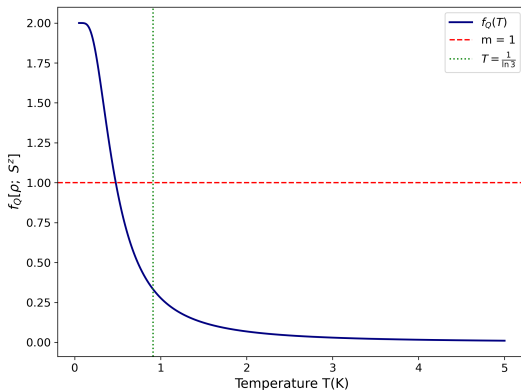


Figure: Analytical f_Q vs T for $N = 2$ at $B = 0$
($\hbar = k_B = 1, J = 1$)



Core Numerical result: $f_Q[\rho; \hat{S}^z](T)$ ($N > 2, B = 0$)

Operational workflow:

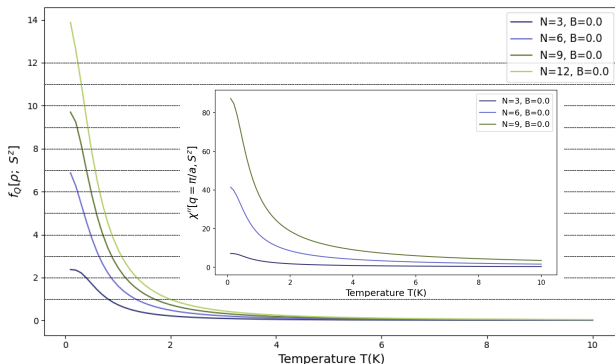
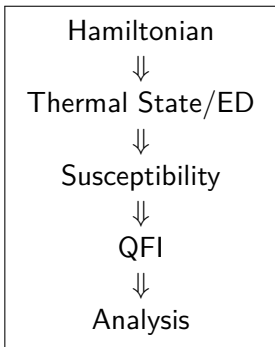


Figure: f_Q vs T at $B = 0$ for varying N ($\hbar = k_B = 1, J = 1$). Dashed horizontal reference lines (for $m \in \mathbb{Z}_+$) indicate different entanglement depth. **Connected correlations $\Rightarrow f_Q$ tracks long-range structure.**



Entanglement scaling with size and temperature

Finite Temperature Regime:

- For $N \ll \xi(T)$: The system behaves as if critical;

$$f_Q \sim \log N$$

- For $N \gg \xi(T)$: Correlations decay $\sim e^{-|i-j|/\xi}$, and QFI saturates.

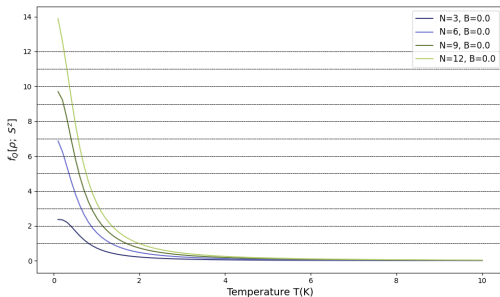


Figure: f_Q vs T at $B = 0$ for different system sizes N . Dashed lines mark entanglement depth thresholds.

Low T: QFI detects increasingly deep multipartite structure as size grows



Crossover Temperature and Entanglement Depth

- Larger $N \rightarrow$ lower thermal window ($\downarrow T_{m/N}$).
- $\uparrow m/N$ (higher entanglement depth) $\Rightarrow \downarrow T_{m/N}$ (lower crossover scale).

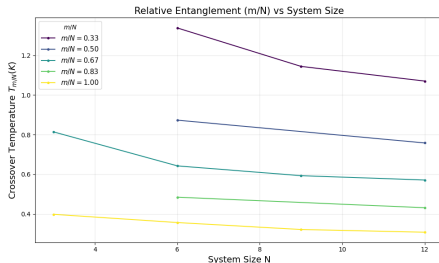


Figure: Crossover temperature $T_{m/N}$ ($T_{f_Q(T)} = m - 1$) vs system size N .



Spin-resolved contributions: where QFI lives

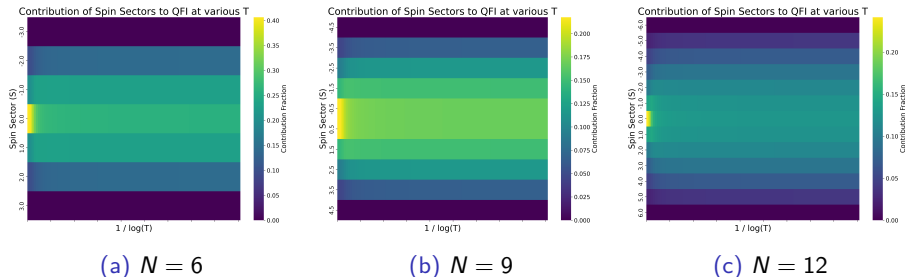


Figure: Heatmaps showing $f_Q(T)$ distribution across total spin sectors S vs T at $B = 0$ for different N .

$T \uparrow \implies$ thermal fluctuation led repopulation, **sector broadening**; however, a substantial fraction of QFI remains confined to low-spin sectors.



Magnetic field suppresses the witness

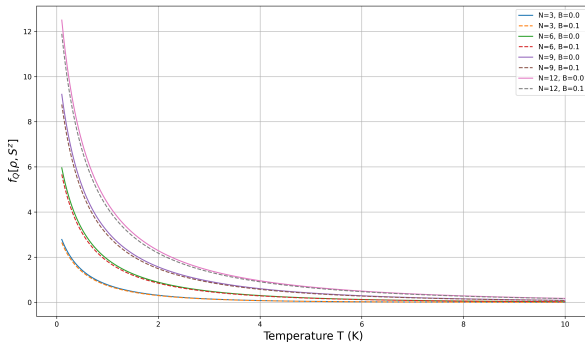


Figure: f_Q vs T for various $B = 0.1$ with $N = 3, 6, 9, 12$ ($\hbar = k_B = 1, J = 1$). The presence of entanglement ($f_Q > 1$) is suppressed by both increasing T and B : field induced spin-rotation symmetry breaking $\Rightarrow$ significantly reduced contributions from symmetric spin sectors.



- QFI can serve as a reasonable operational multipartite entanglement measure for our target system.
- Its signal is strongly shaped by temperature, system size, and applied field.
- **Advantages:** operational interpretation, scalability (using collective measurements), experimental accessibility (estimated from collective spin measurements or interferometric setups [16]).
- It is informative but not exhaustive; conclusions depend on observable choice and witness limitations.



Limitations and caveats

- QFI is sufficient, not necessary.
- **Observable choice (phase generator) matters**[15, 18].
- Witness strength is **not** the same as full entanglement characterization.
- Non-convexity, finite-size numerics and model choices limit generality.
- Sensitivity to noise[4, 6].



Analytical

- $N = 2$ closed-form $f_Q(T)$, benchmarked against PPT/negativity.

Numerical ($N = 3-12$, $B = 0$, $B \neq 0$)

- Two regimes: $f_Q \sim \log N$ (critical) vs. saturation (short-range), set by N vs. $\xi(T)$.
- Crossover temperature $T_{m/N}$: entanglement-depth detection window shrinks with both N and m/N .
- Sector-resolved QFI: entanglement concentrated in low-spin sectors, even as occupation broadens with T .
- Field suppresses f_Q by depleting exactly those sectors ($SU(2) \rightarrow U(1)$).

Scope

- Mapped where QFI works as a witness here, and where it doesn't.



Open directions, future outlook

- Other generators/observables: does a 'suitable' observable always exist?
- Enhanced access to deeper entanglement across systems[3].
- Open-system and dissipative extensions.
- **Real time entanglement tracking**[17].



Thank You!

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Backup slides

Commonly used measures

- ... The most basic of tests: PPT etc.. and what were the results..
- Concurrence:

$$C(\rho) = \max(0, 2\lambda_{\max} - \text{tr}(R)) \quad ; \quad R = \sqrt{\sqrt{\rho}\tilde{\rho}\sqrt{\rho}} \quad (7)$$

where $\lambda_{\max}$ is the maximum eigenvalue of matrix R , and

$$\tilde{\rho} = (\sigma_2 \otimes \sigma_2)\rho^*(\sigma_2 \otimes \sigma_2)$$

- **There is no widely accepted definition of concurrence for systems (larger quantum objects) other than a pair of qubits.**



Commonly used measures

- Tangles:

Configuration	$\tau_1 = 1 - 4 \sum_{\alpha} \langle \mathbf{S}^{\alpha} \rangle^2$	$\tau_2 = 2 \sum_{r \neq 0} C_r^2$
Classical FM	0	0
Classical AFM	0	0
Quantum HF (g.s.)	0	0
Quantum HAF (g.s.)	1	finite

- **Size-based limits on information prevent access to genuine multipartite entanglement.**



Classical Fisher Information I

Let $(X, \mathcal{B}, \mu)$ be a probability space. If $\theta = (\theta_1, \dots, \theta_n)$ is a parameter vector in a neighborhood of $\theta_0 \in \mathbb{R}^n$, then $\exists$ a smooth family μ_θ of probability measures with probability density f_θ :

$$\mu_\theta(H) = \int_H f_\theta(x) d\mu(x), \quad H \in \mathcal{B} \quad (8)$$

The Fisher information matrix at θ_0 is given by:

$$J(\mu_\theta; \theta_0)_{ij} := \int_X f_{\theta_0}(x) \left(\frac{\partial}{\partial \theta_i} \log f_\theta(x) \Big|_{\theta=\theta_0} \right) \left(\frac{\partial}{\partial \theta_j} \log f_\theta(x) \Big|_{\theta=\theta_0} \right) d\mu(x) \quad (9)$$

Alternatively,

$$\mathbf{J}(\mu_\theta; \theta_0)_{ij} = - \int_X \mathbf{f}_{\theta_0}(\mathbf{x}) \frac{\partial^2}{\partial \theta_i \partial \theta_j} \log \mathbf{f}_\theta(\mathbf{x}) \Big|_{\theta=\theta_0} d\mu(\mathbf{x}) \quad ; \quad 1 \leq i, j \leq n \quad (10)$$



Assume that θ is a single parameter. A random variable $\hat{\theta}$ is an unbiased estimator for the parameter θ if:

$$\mathbb{E}_{\theta}(\hat{\theta}) := \int \hat{\theta}(x) f_{\theta}(x) d\mu(x) = \theta, \quad \forall \theta \quad (11)$$

The **Cramér-Rao inequality** states:

$$\text{Var}(\hat{\theta}) := \mathbb{E}_{\theta}((\hat{\theta} - \theta)^2) \geq \frac{1}{\mathbf{J}(\mu_{\theta}; \theta)} \quad (12)$$



For multiple parameters θ_i , the relevant object is given by the so-called quantum Fisher information matrix,

$$F_{ij} = \frac{1}{2} \text{tr}\{(\hat{L}_{\theta_i} \hat{L}_{\theta_j} + \hat{L}_{\theta_j} \hat{L}_{\theta_i}) \rho\} \quad (13)$$

where the symmetric logarithmic derivatives $\hat{L}_{\theta_i}$ satisfy,

$$2\partial_{\theta_i} \rho = \hat{L}_{\theta_i} \rho + \rho \hat{L}_{\theta_i} \quad (14)$$

In Lehmann representation we have,

$$\mathbf{F}_{ij} = 2 \sum_{\mathbf{p}_k + \mathbf{p}_l > 0} \frac{\langle \mathbf{k} | \partial_{\theta_i} \rho | \mathbf{l} \rangle \langle \mathbf{l} | \partial_{\theta_j} \rho | \mathbf{k} \rangle}{\mathbf{p}_k + \mathbf{p}_l} \quad (15)$$



Consider the spectral decomposition of observable A as

$$A = \sum_{i=1}^k \lambda_i E_i$$

On the set $X = \{1, 2, \dots, k\}$, we have probability distributions:

$$\mu_{\theta}(i) = \text{Tr} \rho(\theta) E_i$$

such that,

$$\sum_{i=1}^k \mu_{\theta}(i) = \text{Tr} \rho(\theta) \sum_{i=1}^k E_i = \text{Tr} \rho(\theta) = 1.$$



Since

$$\mu_{\theta}(i) = \frac{\text{Tr} \rho(\theta) E_i}{\text{Tr} D E_i} \cdot \text{Tr} D E_i$$

we can take:

$$\mu(i) = \text{Tr} D E_i$$

where D is a statistical operator. Then,

$$f_{\theta}(i) = \frac{\text{Tr} \rho(\theta) E_i}{\text{Tr} D E_i} \quad (16)$$

Thus, the Fisher information, as defined earlier, can be written as

$$\sum_i \frac{[\text{Tr} \partial_{\theta} \rho(\theta) E_i]^2}{\text{Tr} \rho(\theta) E_i} \quad (17)$$



QFI utilizing measurement III

With spectral decomposition

$$\rho = \sum_k \lambda_k |k\rangle \langle k|,$$

the quantum Fisher information can be written as

$$\mathbf{F}(\rho, \mathbf{A}) = 2 \sum_{\mathbf{k}, \mathbf{l}} \frac{(\lambda_{\mathbf{k}} - \lambda_{\mathbf{l}})^2}{(\lambda_{\mathbf{k}} + \lambda_{\mathbf{l}})} |\langle \mathbf{k} | \mathbf{A} | \mathbf{l} \rangle|^2 \quad (18)$$



For a canonical ensemble, the state of our quantum system is described as

$$\rho = e^{-\beta H} / Z \quad (19)$$

Using Kubo Formula, imaginary part of the response function with respect to $\hat{O}$ can be written as

$$\chi''_{ij}(\omega) = \sum_{\lambda, \lambda'} (p_{\lambda} - p_{\lambda'}) \langle \lambda | \hat{O} | \lambda' \rangle \langle \lambda' | \hat{O} | \lambda \rangle \pi \delta(\omega - E_{\lambda'} + E_{\lambda}) \quad (20)$$

where,

$$p_{\lambda} = e^{-\beta E_{\lambda}} / Z \quad ; \quad Z = \sum_{\lambda} e^{-\beta E_{\lambda}} \quad (21)$$



QFI and correlations II

Now, for a thermal state,

$$\frac{p_\lambda - p_{\lambda'}}{p_\lambda + p_{\lambda'}} = \frac{e^{-\beta E_\lambda} - e^{-\beta E_{\lambda'}}}{e^{-\beta E_\lambda} + e^{-\beta E_{\lambda'}}} = \frac{1 - e^{-\beta(E_{\lambda'} - E_\lambda)}}{1 + e^{-\beta(E_{\lambda'} - E_\lambda)}} = \tanh\left(\frac{\beta(E_{\lambda'} - E_\lambda)}{2}\right)$$

We can rewrite (with $\hbar = 1$),

$$\frac{p_\lambda - p_{\lambda'}}{p_\lambda + p_{\lambda'}} = \tanh\left(\frac{\beta(E_{\lambda'} - E_\lambda)}{2}\right) = \int_{-\infty}^{\infty} d\omega \tanh(\omega/2T) \delta(\omega - E_{\lambda'} + E_\lambda)$$

Now, we know that for a mixed state $\rho = \sum_I p_I |I\rangle\langle I|$, $F_Q[\rho; \hat{O}]$ is given by,

$$F_Q[\rho; \hat{O}] = 2 \sum_{I, I'} \frac{(p_I - p_{I'})^2}{p_I + p_{I'}} |\langle I | \hat{O} | I' \rangle|^2 \quad (22)$$

i.e.,

$$F_Q[\rho; \hat{O}] = 2 \sum_{I, I'} \frac{p_I - p_{I'}}{p_I + p_{I'}} (p_I - p_{I'}) |\langle I | \hat{O} | I' \rangle|^2 \quad (23)$$



If the mixed state is a thermal state, we can write,

$$F_Q[\rho; \hat{O}] = 2 \sum_{l,l'} \int_{-\infty}^{\infty} d\omega \tanh\left(\frac{\beta\omega}{2}\right) \delta(\omega - E_{l'} + E_l) (p_l - p_{l'}) |\langle l | \hat{O} | l' \rangle|^2$$

After including appropriate factors, we get,

$$\mathbf{F}_Q[\rho; \hat{O}](\mathbf{T}) = \frac{4}{\pi} \int_0^{\infty} d\omega \tanh\left(\frac{\beta\omega}{2}\right) \sum_{i,j} \mathbf{e}^{i\mathbf{q}(\mathbf{r}_i - \mathbf{r}_j)} \chi''_{ij}(\omega, \mathbf{T}) \quad (24)$$



Numerical results: $f_Q[\rho; \hat{S}^z](T)$ ($N > 2, B = 0$)

Low-Temperature Regime ($T \rightarrow 0$):

The QFI density simplifies to:

$$f_Q = \frac{F_Q}{N} \sim \frac{1}{N} \sum_{i,j} \langle 0 | S_i^z S_j^z | 0 \rangle_c$$

where the connected correlation function

$$\langle S_i^z S_j^z \rangle_c = \langle S_i^z S_j^z \rangle - \langle S_i^z \rangle \langle S_j^z \rangle.$$

In particular, $\langle S_i^z S_j^z \rangle \sim |i - j|^{-\eta}$ for critical models. Thus, we get a scaling:

$$f_Q \sim \log N$$

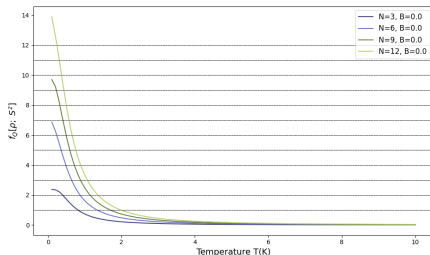


Figure: f_Q vs T at $B = 0$ for different system sizes N . Dashed lines mark entanglement depth thresholds.



Temperature dependence and size scaling:

The 1D isotropic HAF model is characterized by:

- No energy gap ($\Delta = 0$)
- Algebraically decaying spin-spin correlations

Using QFI with local operators $\hat{O}_i = S_i^z$ we have:

- Low T : long-range entanglement via coherent local correlations.
- High T : thermal noise suppression $\implies \text{QFI} \downarrow \implies$ destroyed entanglement.



Temperature dependence and size scaling:

As $N \rightarrow \infty$, $T_{m/N} \rightarrow 0$ for any fixed m/N (no finite- T long-range entanglement in $1D$ gapless systems [10]). For large systems:

- Only short-range multipartite entanglement survives at finite T .
- Detecting genuine global entanglement requires cooling to very low T (may not be experimentally feasible).



Comparison to other systems:

- *Gapped Systems:* In systems with a finite gap $\Delta > 0$,

$$T_{m/N} \rightarrow T_c(m/N) > 0$$

Here, multipartite entanglement can survive up to a finite critical temperature T_c [8].

- *Higher-Dimensional Systems:* In higher dimensions (e.g., 2D HAF), critical behavior can differ:
 - Longer correlation lengths
 - Possibility of quasi-long-range or true long-range order
 - Slower decay of $T_{m/N}$ as N increases



Comparison to other systems:

- *Topological Systems*: In topologically ordered systems (e.g., toric code, Kitaev chain), entanglement may persist to finite temperature:

$$T_{m/N} \rightarrow T_{\text{topo}}(m/N)$$

However, the QFI may not directly reflect topological order, depending on the observable used [1].



Role of spin sectors:

- Preferred spin configurations for high entanglement.
- Sector broadening: thermal fluctuations erode entanglement by mixing in symmetry sectors that are less quantum-correlated [14].
- Use of QFI: in systems with conserved quantum numbers—such as total spin, particle number, or angular momentum—symmetry-resolved QFI can be used to dissect entanglement dynamics.



Effect of Magnetic Field:

- Field-induced suppression of QFI: local/global symmetries significant to preserving entanglement.
- QFI is a symmetry-sensitive witness in both equilibrium and non-equilibrium settings for field sensitive systems.



System Dependence: Type, Size, and Symmetries

- *Spin systems*: Systems of qubits (spin-1/2 particles), spin-squeezed states, Dicke states, and GHZ states [13].
- *Bosonic systems*: Cold atoms in BECs (same spatial mode, collective spin dynamics), spin-squeezed states produced via nonlinear interactions [7].
- *Photonic systems*: Entangled photons in interferometry, especially NOON states and other path-entangled states [5].
- *Hybrid and continuous-variable systems*: QFI applies with modifications to accommodate non-linear or infinite-dim. generators [9].



Usefulness Across Systems

- **Advantages:** Operational Interpretation, Scalability (using collective measurements), Experimental Accessibility (estimated from collective spin measurements or interferometric setups [16]).
- **Limitations:** Sufficiency but not necessity and limited scope, dependence on generator choice, non-convexity, sensitivity to noise[4, 6].

